

Verify Important Outputs

Project: Helpful Classroom Sorter

1. Your mission

Your learning target

I can separate an answer into claims and connect each claim to relevant evidence.

I can recalculate quantities with consistent units and a reasonableness check.

I can report supported, contradicted and unresolved claims without inventing missing evidence.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Prompts Are Design Instructions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Good Data, Bad Data. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Fluent AI output is not evidence; important claims need independent support.



LEARNING TARGET A generated report says 18 of 24 tests passed, so success was 80%. Verify the claim in two ways and write a corrected sentence.

Learner name:

Readiness for the connected explanation

A response follows the requested four-line format but invents a missing start time. Has it passed the task? Explain.

This week's end goal

By the end, I can verify quantities with consistent units and separate arithmetic from unsupported claims.

2. Notice the evidence

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; evidence tells us what the clues mean.



MYSTERY A generated paragraph confidently claims that an animal lives in a region; the wording is not evidence that the claim is true.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first prediction and state the clue that supports it.

My observations, prediction, and evidence:

GOOD THINKING A prediction can change when stronger evidence appears.

A closer explanation

An answer can contain a mixture of correct, incorrect and unverified statements. Checking one sentence does not validate every other sentence. First separate the answer into individual claims. Then ask what evidence would support each one: a record, a measurement, a calculation or some combination.

Our fictional ribbon activity uses current materials card R. R states: four rolls of ribbon, each exactly 2 metres long; twelve pieces will each use 50 centimetres; no cutting loss is included in this paper model. One metre equals 100 centimetres. An old card Q describes a different activity with three rolls. These are provided classroom records, not instructions to cut real materials.

A draft says “R gives four rolls. The starting length is 8 metres. The pieces use 600 metres. Two metres remain. Every participant will enjoy the activity.” Do not approve or reject the whole paragraph at once.

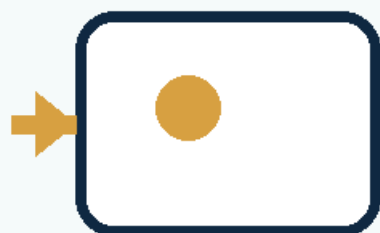
Inspect the roll count, starting length, used length, remaining length and enjoyment claim separately.

3. See how it works

VISUAL MECHANISM

Input → **Process** → **Output**

Follow the information instead of imagining magic inside the machine.



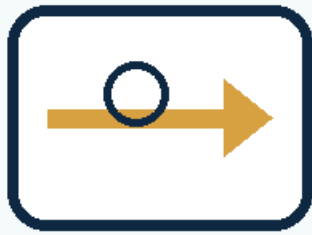
INPUT

specific claim extracted from an output



PROCESS

check independent suitable sources and
reconcile differences



OUTPUT

supported, corrected, uncertain, or removed claim

INPUT specific claim extracted from an output

PROCESS check independent suitable sources and reconcile differences

OUTPUT supported, corrected, uncertain, or removed claim

CORE IDEA Fluent AI output is not evidence; important claims need independent support.

UNDER THE HOOD Verification is claim-level evidence work. Source independence, authority, direct support, recency, and agreement matter. Multiple pages repeating the same upstream error are not independent confirmation.

Draw a fresh example with all three stages:

A closer explanation

Different claims need different checks

Claim · Evidence path · Decision

Claim: Four rolls

Evidence path: Read current card R

Decision: Supported

Claim: 8 metres at start

Evidence path: 4 rolls $\times$ 2 metres

Decision: Supported

Claim: 600 metres used

Evidence path: $12 \times 50 \text{ cm} = 600 \text{ cm}$

Decision: Wrong unit

Claim: Everyone will enjoy it

Evidence path: No enjoyment evidence

Decision: Unresolved

A correct source reference is a starting point. Read what the record actually says and check every derived quantity.

A citation is useful only if you can locate the source and confirm that it supports the specific claim. The title “Materials card R” beside a paragraph does not make the enjoyment claim part of R. If a cited card Z is not supplied or cannot be found, that citation cannot currently verify its claim. Record the gap rather than inventing Z’s contents.

Check whether the source covers the right activity, version, population or time. Q’s three rolls may be accurate for its own activity but irrelevant to the current four-roll task. In real work, also inspect who created the record, how they obtained the information and whether there is a correction or better source.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

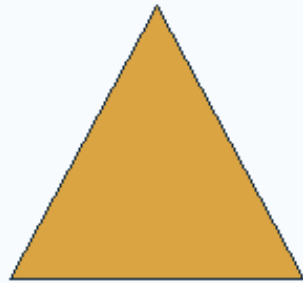
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the evidence carefully.



Generated sentence



Zoo care page



Two copied blogs



Conflicting references

CASE 1 — Generated sentence Clue: fluent output. Meaning: claim, not evidence.

CASE 2 — Zoo care page Clue: direct animal information. Meaning: possible source.

CASE 3 — Two copied blogs Clue: same origin. Meaning: not independent.

CASE 4 — Conflicting references Clue: evidence disagrees. Meaning: report uncertainty.

BOUNDARY Two search results are not two independent sources when both copy the same page.

Which two pictures are easiest to confuse? What evidence separates them?

Change one thing in the pictured case

Change: Make ten bookmarks instead of eight.

Prediction: 100 cm, or 1 m, remains.

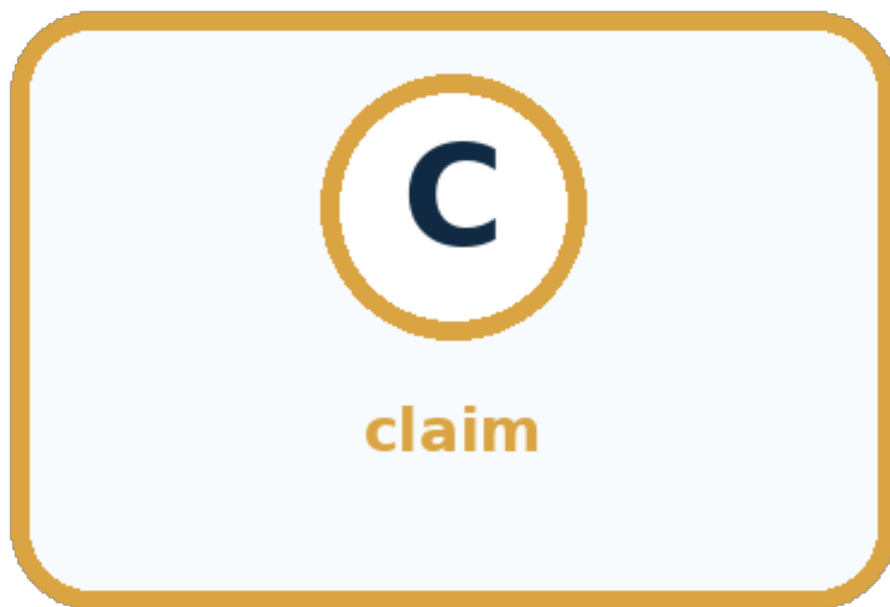
Reason: $600 - 10 \times 50 = 100$ cm.

5. Words that unlock the idea

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.





source



independent



Word · Meaning · My example

Word: claim

Meaning: statement that can be checked

My example: _____

Word: source

Meaning: origin of evidence

My example: _____

Word: independent

Meaning: not merely copying the same origin

My example: _____

Word: corroborate

Meaning: support a claim with separate evidence

My example: _____

Misconception detective

CLAIM TO REPAIR Professional wording is evidence of accuracy.

Use one vocabulary word and one piece of evidence to repair it:

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word · Meaning

Word: Verification

Meaning: Checking whether a claim is supported by appropriate evidence

Word: Citation

Meaning: A reference that helps a reader locate a source

Word: Unit

Meaning: The agreed measurement scale attached to a quantity, such as centimetres

Word: Corroboration

Meaning: Additional evidence that supports a claim, with its origin checked

6. Follow a worked example

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable evidence.



CASE A generated paragraph confidently claims that an animal lives in a region; the wording is not evidence that the claim is true.

STEP 1 — OBSERVE specific claim extracted from an output

STEP 2 — TRACE check independent suitable sources and reconcile differences

STEP 3 — CONCLUDE supported, corrected, uncertain, or removed claim

MODEL REASONING Check the exact claim, not the whole paragraph. Use sources with direct relevant evidence, note publication context, and explain whether they independently agree. If evidence is weak or conflicting, revise or remove the claim.

DO NOT OVERCLAIM Two search results are not two independent sources when both copy the same page.

What new evidence could change this answer?

A closer explanation

Quantity · Calculation in centimetres · Equivalent in metres

Quantity: Starting length

Calculation in centimetres: $4 \times 200 \text{ cm} = 800 \text{ cm}$

Equivalent in metres: 8 m

Quantity: Length used

Calculation in centimetres: $12 \times 50 \text{ cm} = 600 \text{ cm}$

Equivalent in metres: 6 m

Quantity: Length remaining

Calculation in centimetres: $800 \text{ cm} - 600 \text{ cm} = 200 \text{ cm}$

Equivalent in metres: 2 m

Convert to one length unit before subtracting. Two metres is 200 centimetres, so four rolls give 800 centimetres. The twelve pieces use 600 centimetres, leaving 200 centimetres, or 2 metres. Subtracting the bare numbers $8 - 600$ would mix metres with centimetres and would not describe the remaining length correctly.

Use a second route as a check: 50 centimetres is half a metre, so twelve pieces use six metres. Eight minus six leaves two metres. A reasonableness check also asks whether the used length is at most the starting length and whether starting length equals used plus remaining. These checks can reveal mistakes; they do not replace checking the original inputs and assumptions.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Verify Important Outputs

A fictional craft sheet calculates ribbon left after making bookmarks.

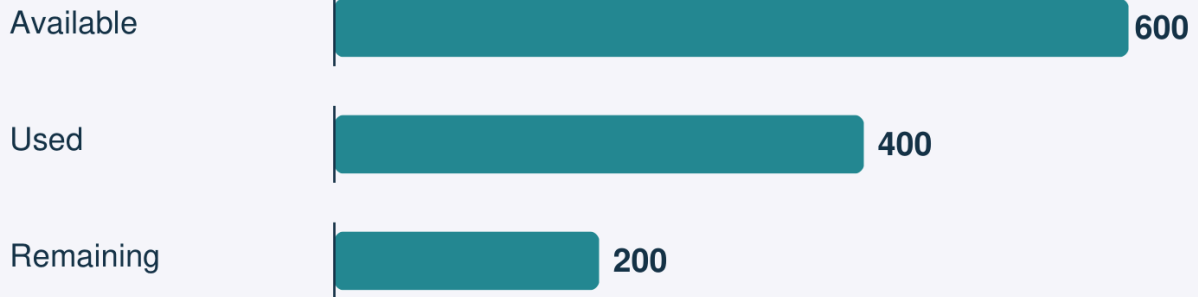
The facts we will use

Four rolls each contain exactly 150 cm. Eight bookmarks use exactly 50 cm each.

No other ribbon loss is assumed; no competition result is recorded.

Ribbon quantities in one consistent unit

centimetres



Bar length shows the supplied value; the marked baseline is zero.

Verify Important Outputs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Available ribbon

$$4 \times 150 = 600 \text{ cm.}$$



2. Ribbon used

$$8 \times 50 = 400 \text{ cm.}$$



3. Remainder

$$600 - 400 = 200 \text{ cm} = 2 \text{ m.}$$



4. Audit draft

"200 m remains and everyone wins" fails unit and evidence checks.

CHANGE ONE THING

Make ten bookmarks instead of eight.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Available ribbon

What happens and why: $4 \times 150 = 600 \text{ cm.}$

Step: 2. Ribbon used

What happens and why: $8 \times 50 = 400 \text{ cm.}$

Step: 3. Remainder

What happens and why: $600 - 400 = 200$ cm = 2 m.

Step: 4. Audit draft

What happens and why: “200 m remains and everyone wins” fails unit and evidence checks.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

All arithmetic first uses centimetres. Converting only after subtraction avoids mixing units. The record supports available, used and remaining lengths under its no-loss assumption. It does not support a promise about winning. A numerically correct total elsewhere in a draft cannot rescue a wrong unit or an invented claim.

What this example does not establish

Real cutting losses or inaccurate roll lengths would change this toy calculation; the assumption must be declared.

Read the labelled picture: Correct the draft and identify which part needs another source rather than a calculation.

Change one thing: Make ten bookmarks instead of eight. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case · Evidence · What it shows

Case: Generated sentence

Evidence: fluent output

What it shows: claim, not evidence

Case: Zoo care page

Evidence: direct animal information

What it shows: possible source

Case: Two copied blogs

Evidence: same origin

What it shows: not independent

Case: Conflicting references

Evidence: evidence disagrees

What it shows: report uncertainty

Repair two tempting claims

CLAIM A Professional wording is evidence of accuracy.

Repair:

CLAIM B The first two agreeing links settle every claim.

Repair:

STRONG ANSWER FRAME I claim _____ because the evidence shows _____. The process is _____. One limit is _____.

A closer explanation

A useful correction names the original claim, relevant source, calculation with units and revised conclusion. For example: R lists four 2-metre rolls and twelve 50-centimetre pieces. The pieces use 600 centimetres, or 6 metres; the paper model leaves 2 metres when no cutting loss is included. This preserves the condition on which the result depends.

If the real activity loses ribbon at each cut, the no-loss result cannot simply be reused. Obtain that missing information or label the answer as a model with an explicit assumption. Similarly, if you cannot access a source, mark the claim unresolved and identify what evidence would settle it. Repeating a confident answer is not a new measurement.

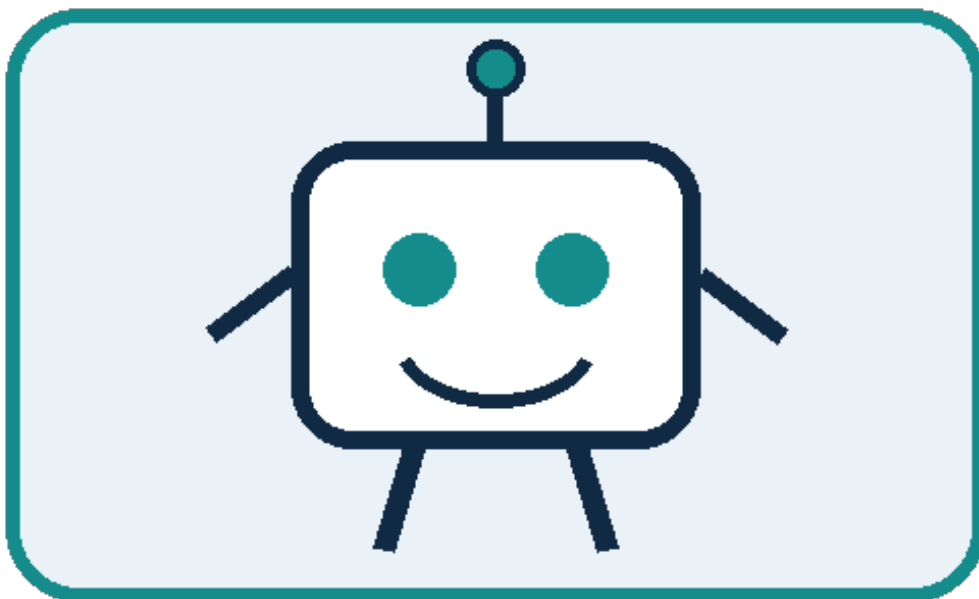
In a changed paper model, each of the twelve pieces uses 50 cm and an extra 2 cm is lost for each piece. Calculate the total loss and the remaining ribbon from the same four rolls. State which assumption changed

8. Practise together

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Fluent AI output is not evidence; important claims need independent support.

Use a fresh example to explain claim.

Trace a new input → process → output example.

Find and repair the error in: The first two agreeing links settle every claim.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the evidence. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

The claim “600 metres used” conflicts with the stated inputs and calculation. The claim “everyone will enjoy it” is not supported by R or by any supplied enjoyment evidence. That does not establish that nobody will enjoy it. State “not verified from these records” rather than changing an unsupported positive claim into an unsupported negative claim.

Audit the draft from section one. For each of its five claims, state supported, contradicted or unresolved under the supplied evidence. Correct the used-length claim and explain why the correct remaining-length claim still needs its own calculation

Suppose three posters repeat the same number and each copied it from one unverified draft. That is three copies of one claim, not three independent measurements. Follow references back to their origin. A separate check of R and the arithmetic provides more useful evidence than counting copies that repeat the same mistake.

Practise with less help: Repair a unit error

Use workbook W2 for the connected example “Verifying claims and quantities”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Verifying claims and quantities. Proposed explanation: “A citation proves every nearby claim”. Decide whether this explanation is supported. Point to the step or supplied fact that decides;

explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Correct the draft and identify which part needs another source rather than a calculation.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EXPLORER LAB

Two-Source Claim Check

Predict first, test honestly, and explain what the evidence means.



State your prediction before the result is known.

Complete the task: Use a two-source checklist on a generated animal fact.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

Case · Prediction · Observed result · What it means

Case: Normal

Prediction: _____

Observed result: _____

What it means: _____

Case: Edge

Prediction: _____

Observed result: _____

What it means: _____

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

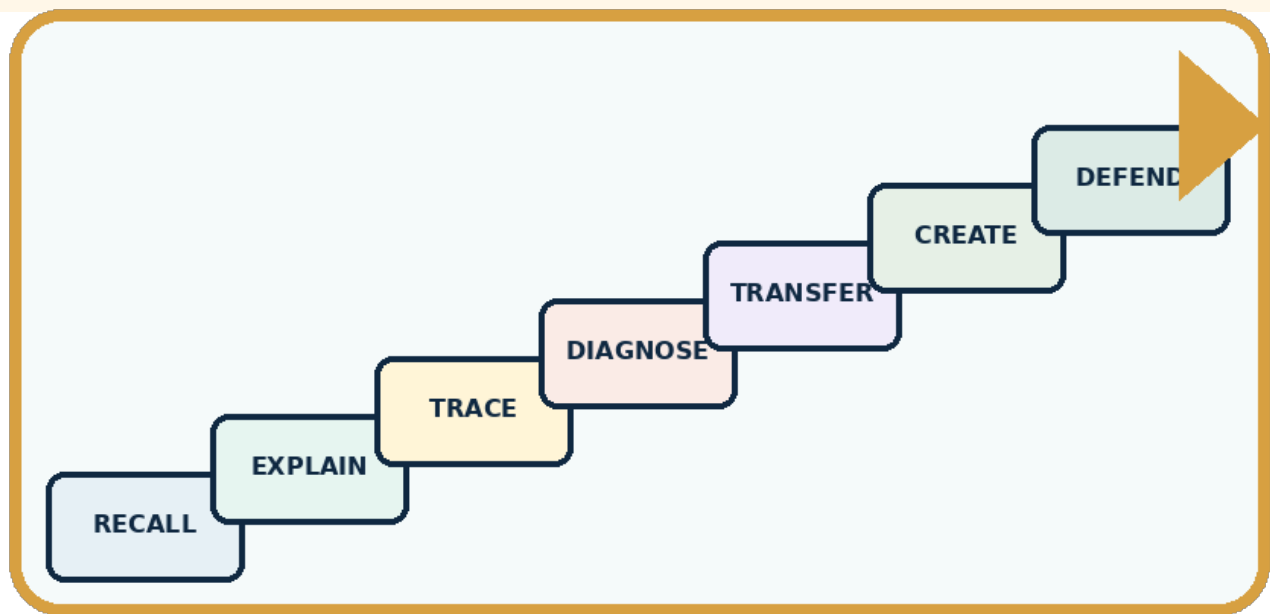
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define claim.

EXPLAIN Fluent AI output is not evidence; important claims need independent support.

TRACE Show the input, process, and output.

DIAGNOSE Repair: Professional wording is evidence of accuracy.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, evidence, mechanism, limit, and safe action.

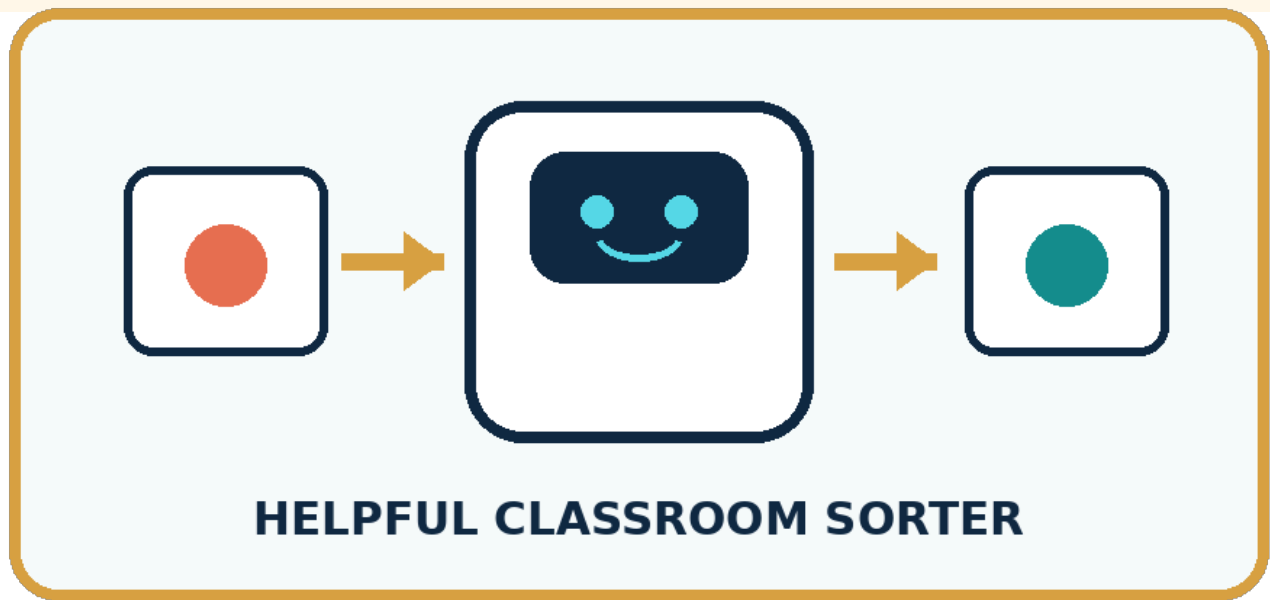
RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

11. Connect your project

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one evidence-based decision at a time.



THIS WEEK'S CHECKPOINT Require human confirmation for important sorter outputs and attach the evidence record used to approve each rule or label.

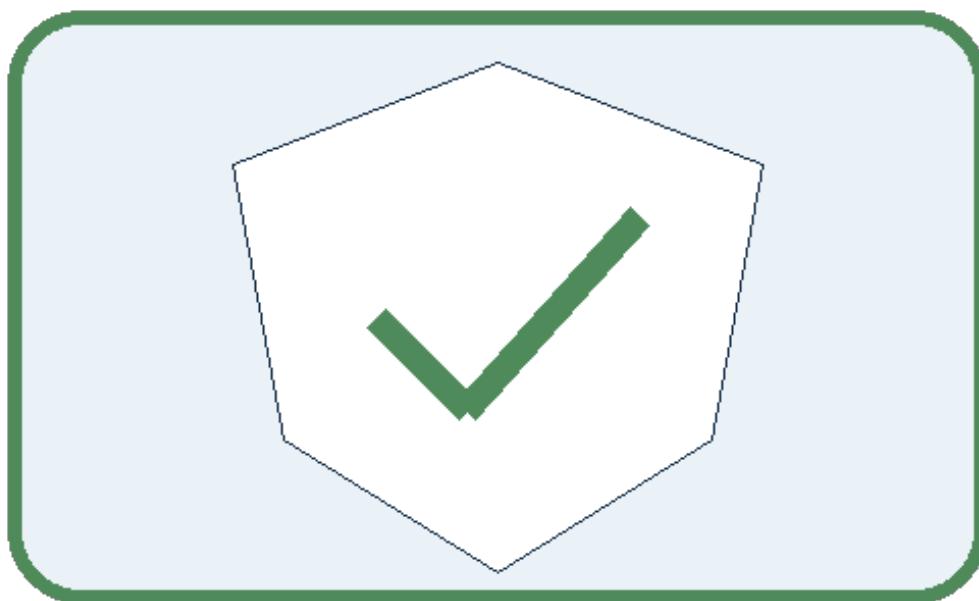
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the process?

What normal and edge cases will you test?

When should it say not sure or ask a person?



RESPONSIBLE DESIGN Two search results are not two independent sources when both copy the same page.

My project decision:

Evidence for your project

Check sorter calculations with explicit units and verify explanation claims against their actual catalogue evidence.

12. Show what you know

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



REQUIRED PROOF Extract a generated animal claim, verify it with two independent sources, and record support, correction, or uncertainty.

- ☐ Name or draw the fresh case.
- ☐ Trace input, process, and output.
- ☐ State the conclusion and strongest evidence.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.

☐ Explain one safe next action or improvement.

Independent evidence:

SELF-CHECK Check the exact claim, not the whole paragraph. Use sources with direct relevant evidence, note publication context, and explain whether they independently agree. If evidence is weak or conflicting, revise or remove the claim.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

NEXT BRIDGE Carry today's evidence habit into: Humans Set Boundaries.

13. Study a complete case

Week 27 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

An answer says six groups of four make 28.

The question

Verify independently.

Worked explanation

Six times four is 24. Draw six groups or add four six times. Confident wording does not change arithmetic.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Six groups of four make twenty-four

Evidence or stage · Value or result

Evidence or stage: Group 1

Value or result: 4

Evidence or stage: Group 2

Value or result: 4

Evidence or stage: Group 3

Value or result: 4

Evidence or stage: Group 4

Value or result: 4

Evidence or stage: Group 5

Value or result: 4

Evidence or stage: Group 6

Value or result: 4

An answer says six groups of four make 28.

Verify independently.

Six times four is 24. Draw six groups or add four six times. Confident wording does not change arithmetic.

14. Try a new case independently

Independent case: Audit a new model with mass units

Fictional craft card M states five sealed bags of modelling material at exactly 250 g per bag. Three models each use 300 g, with no other loss. A draft says “M supplies 1250 g; the models use 900 g; 350 kg

remains; every model will win a prize.” Check all four claims, show the remaining mass in grams, and explain the status of the prize claim. Use $1\text{ kg} = 1000\text{ g}$.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 27 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

An assistant says the school library opens at 8, but today's official notice says 9. Which source should guide today's visit? Explain the check.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Require human confirmation for important sorter outputs and attach the evidence record used to approve each rule or label.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A generated report says 18 of 24 tests passed, so success was 80%. Verify the claim in two ways and write a corrected sentence.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional craft sheet calculates ribbon left after making bookmarks.

By the end, I can · How I will show it

By the end, I can: Separate an answer into claims and connect each claim to relevant evidence

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Recalculate quantities with consistent units and a reasonableness check

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Report supported, contradicted and unresolved claims without inventing missing evidence

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Separate an answer into claims and connect each claim to relevant evidence

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional craft sheet calculates ribbon left after making bookmarks.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? “200 m remains and everyone wins” fails unit and evidence checks.

1. Audit draft
2. Remainder
3. Ribbon used
4. Available ribbon

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Audit draft “200 m remains and everyone wins” fails unit and evidence checks. All arithmetic first uses centimetres. Converting only after subtraction avoids mixing units. The record supports available, used and remaining lengths under its no-loss assumption. It does not support a promise about winning. A numerically correct total elsewhere in a draft cannot rescue a wrong unit or an invented claim.

Practice 2: explain the deciding evidence

Changed-case practice: Make ten bookmarks instead of eight. Which conclusion is supported by the

supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. 100 cm, or 1 m, remains.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: 100 cm, or 1 m, remains. $600 - 10 \times 50 = 100$ cm.

Repair a missing building block

Review Prompts Are Design Instructions (ai-1/26). Goal: Write a prompt with a clear task, supplied evidence and checkable output requirements. Short practice: What must happen if B instead conflicts with another equally authoritative duration record?. Return to this week and explain what you changed.

Review Privacy by Choosing Less (ai-1/24). Goal: Justify which information a stated task needs and remove unnecessary fields. Short practice: For the original task, does adding an anonymous-looking pupil code improve the repair count?. Return to this week and explain what you changed.

Review Good Data, Bad Data (ai-1/21). Goal: Check records against a stated format, unit and meaning before using them. Short practice: Which repairs are evidence-backed in the original case, and which row must remain unknown?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.